

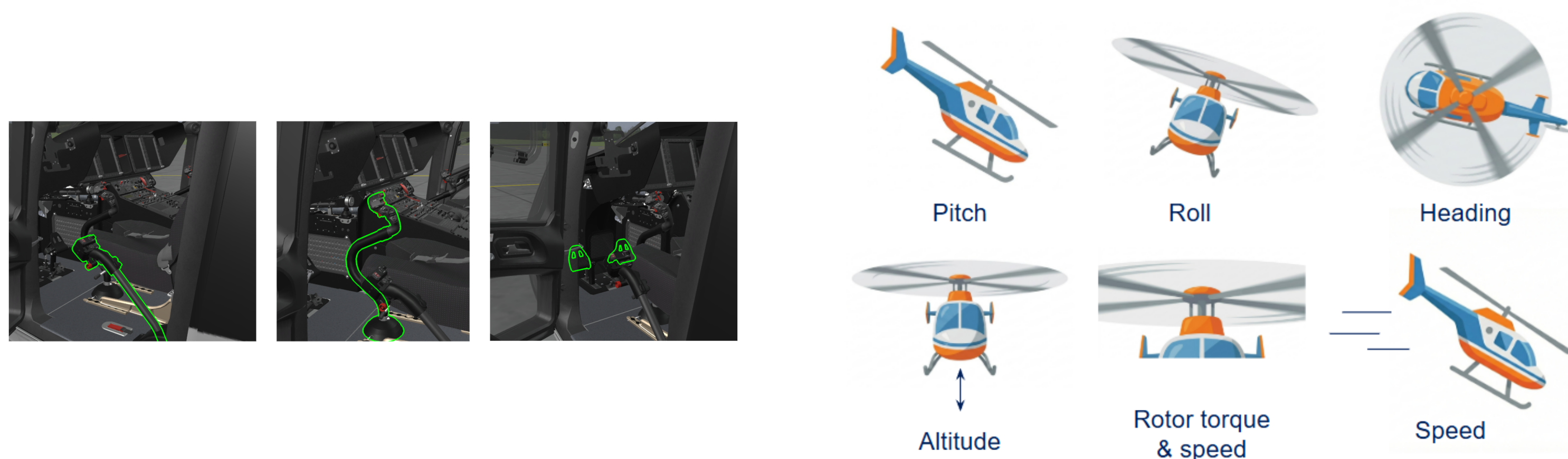
Alignment of Control Inputs for Helicopter Trajectory Simulation: a RL-based Approach

Melvine Nargeot^{1,2,3}, Filippo S. Perotto², Gauthier Picard², Cédric Chauvelon¹, Christine Allain¹

¹ Airbus Helicopters, Marignane - ² DTIS, ONERA, Université de Toulouse - ³ ISAE/SUPAERO, Toulouse

Context

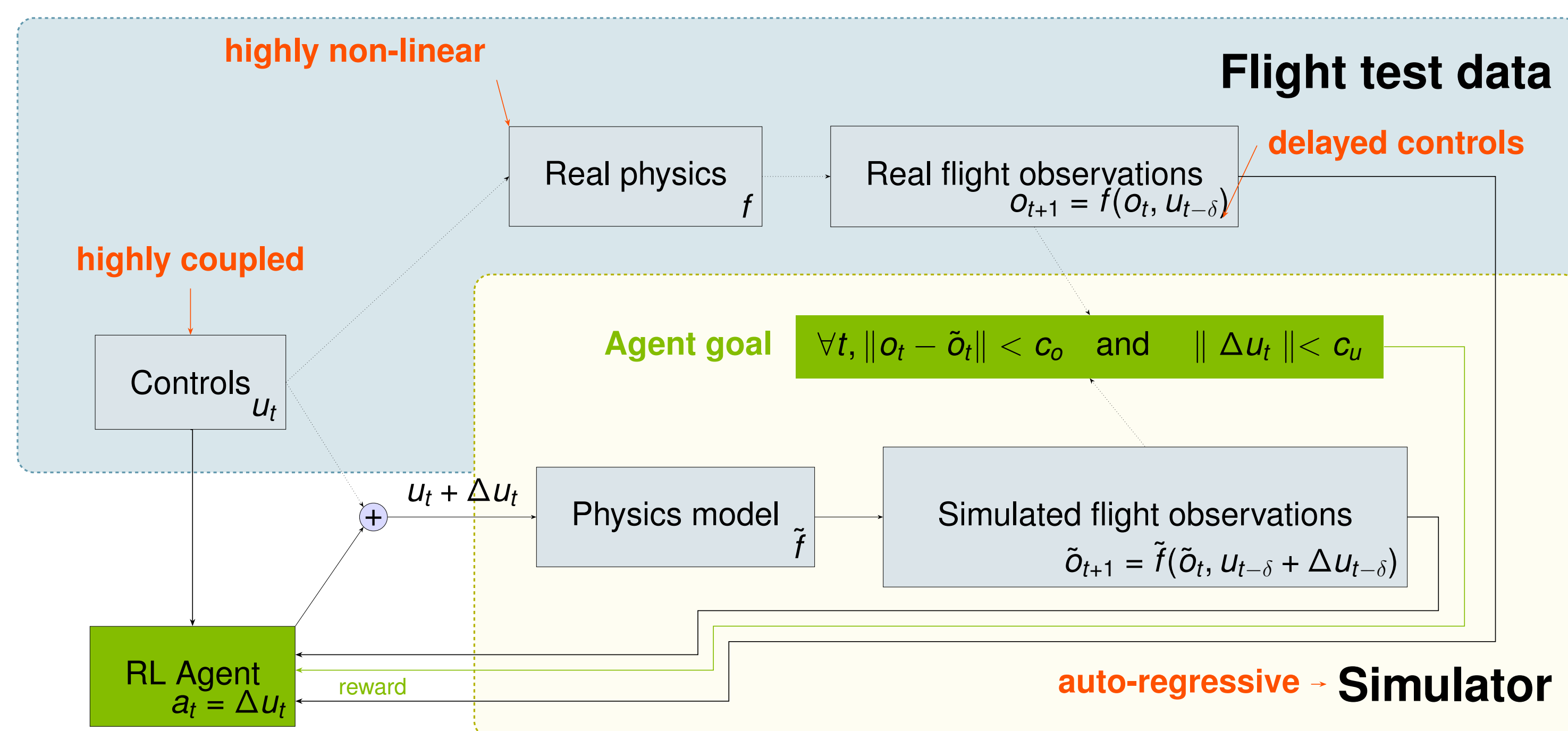
- Certifying helicopter flight simulator requires **accurate trajectory replication**
 - Physical approximations cause rapid **model divergence**
 - Subtle **command corrections** are allowed to realign the trajectories
- **Research question**: How to automate corrections using deep RL?



Challenges

- Auto-regressive simulators → approximation **errors accumulate over time**
- Physics is **non-linear, highly coupled** and commands are **delayed**
- Without corrections, trajectories **exceed certification bounds**
- **Multi-objective optimization**, fitting observations while minimizing control corrections

Model



Experimental Setup

- Custom environment with **abstract transfer functions** ($f, \tilde{f}$)
- 2 inputs, 2 responses
- Train dataset: 10000 trajectories, 500 timesteps each
- Evaluated algorithms: PID, PPO [4], SAC [1]

References

- [1] Tuomas Haarnoja and al. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 1861–1870, 2018.
- [2] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, pages 1–7, 2025.
- [3] Nicklas A Hansen, Hao Su, and Xiaolong Wang. Temporal difference learning for model predictive control. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 8387–8406. PMLR, 2022.
- [4] John Schulman and al. Proximal policy optimization algorithms. *arXiv*, 2017.

Results

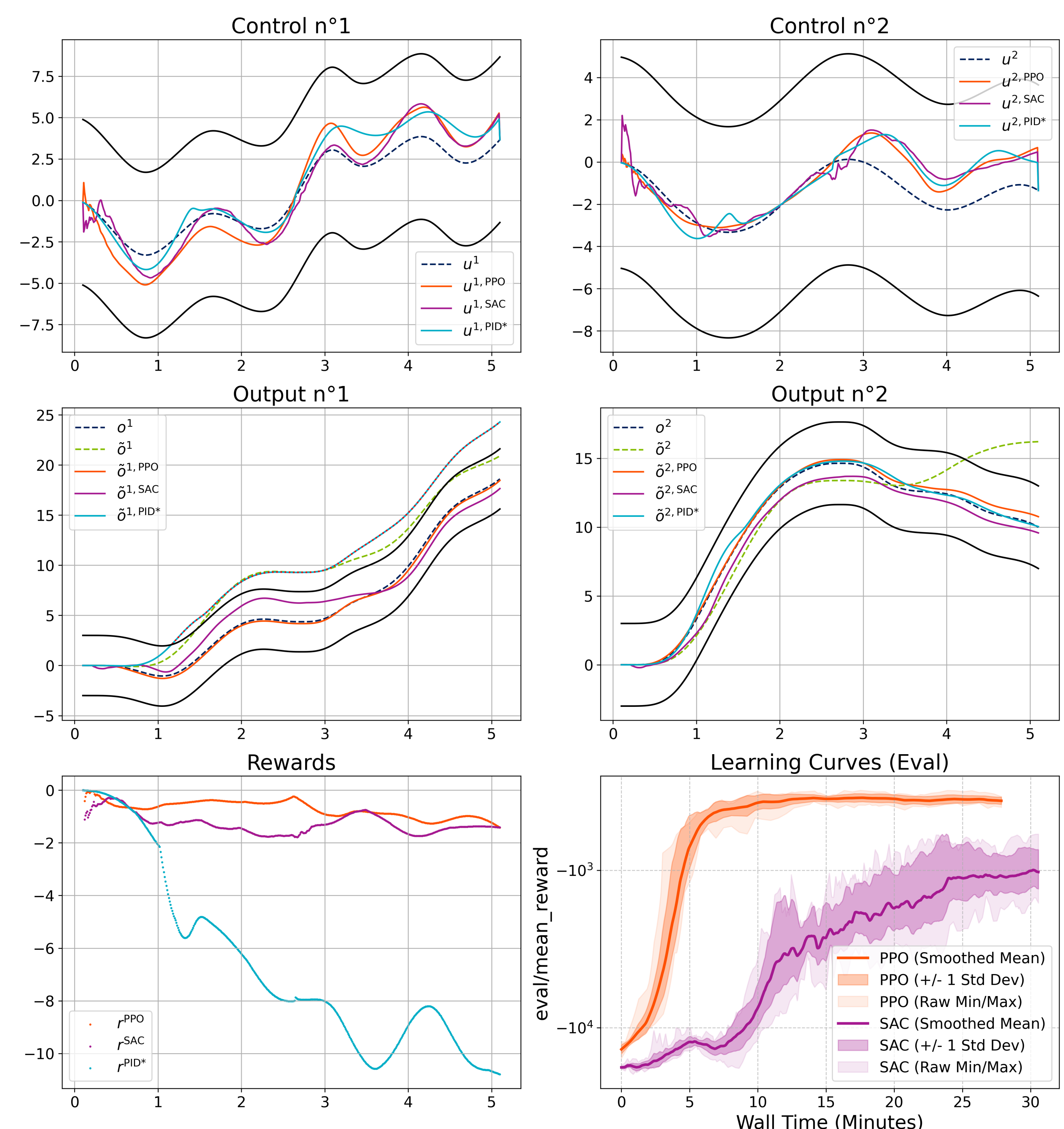


Table 1: Performance comparison of PID*, PPO, and SAC algorithms across different MIMO dimensions. Bold values indicate the best performance for each metric on a set of 30 test trajectories.

Size	RMSE(u)	RMSE(o)	Action Penalty	Constraint Violation
2x2				
PID*	0.80 ± 0.80	1.45 ± 1.84	0.67 ± 1.19	18.10% ± 26.20%
PPO	0.51 ± 0.27	0.84 ± 0.53	0.01 ± 0.00	8.67% ± 13.84%
SAC	0.73 ± 0.49	0.96 ± 0.47	0.01 ± 0.00	10.09% ± 15.88%
3x3				
PID*	0.63 ± 0.45	2.96 ± 2.63	0.01 ± 0.01	37.15% ± 25.26%
PPO	0.72 ± 0.31	1.19 ± 1.24	0.01 ± 0.00	13.17% ± 20.00%
SAC	0.81 ± 0.41	1.56 ± 1.82	0.01 ± 0.00	20.13% ± 22.33%
4x4				
PID*	0.58 ± 0.64	3.15 ± 2.84	0.05 ± 0.13	49.98% ± 18.73%
PPO	0.83 ± 0.38	1.26 ± 1.58	0.02 ± 0.01	15.89% ± 22.70%
SAC	0.44 ± 0.15	2.22 ± 1.25	0.00 ± 0.00	47.57% ± 14.86%

Conclusion & Perspectives

- **Deep RL Superiority**: PPO/SAC counter delays, keeping errors within **regulatory tolerances** via **predictive sliding windows**
- **Industrial Scaling**: Testing policies on **Airbus Helicopters** high-fidelity simulators
- **Safe RL**: Integrating **Control Barrier Functions (CBF)** or **Lyapunov constraints** for strict safety guarantees
- **Sample Efficiency**: Exploring **model-based RL** (*DreamerV3* [2], *TD-MPC2* [3]) to **reduce simulation costs**